



Doubao II

The Doubao System and the UDOS Reasoning Engine: A Technical Comparison of Industrial Productization and a Falsifiable Governance Substrate

豆包体系与 UDOS 推演引擎技术对比



The Doubao System and the UDOS Reasoning Engine: A Technical Comparison of Industrial Productization and a Falsifiable Governance Substrate

One-sentence conclusion: Doubao is a **mature commercial Agent system already running on real users and real workloads**, strong in end-to-end engineering, protocol ecosystem and commercial closed loop, weak in "numbers not independently reproducible, multi-Agent division of labor not documented, lacking formalized governance"; UDOS is a **falsifiable governance substrate still at the CPU prototype stage**, strong in Byzantine fault tolerance, handoff contracts, quality saturation law and three-level evidence discipline, weak in having no product, no real LLM long chains, no edge-side or commercial landing. **The two are fundamentally not at the same maturity—this comparison is not about who wins, but about what the "industrial productization" and "research falsifiability" routes each do thoroughly, and what each lacks.**

Evidence caliber: Doubao side strictly guards three states—[official statement] (official release/press conference/official docs), [media says] (authoritative media or press conference retelling), [to verify] (no official original obtained, not treated as fact); UDOS side strictly guards three evidence levels—**verified** (deterministic property tests/passed CI), **cpu-proto** (CPU prototype single-seed pilot point estimate, rerunnable in `reports7/*.json`), **unverified** (design proposal/direction). All single-seed results uniformly marked "pilot point estimate."

Abstract

This paper places the commercial Agent system of Doubao (ByteDance) and the author's self-developed UDOS Reasoning Engine on the same comparison table, doing one **fair, evidence-grounded, neither hyped nor disparaged** technical contrast. Doubao represents the "industrial productization" route: a five-layer vertically integrated system from model substrate (Doubao 2.1 Pro/Turbo, UI-TARS GUI substrate), protocol layer (MCP/A2A/SAEP), development and enterprise platforms (Coze/TRAE/AgentKit/HiAgent/ArkClaw/AI Trust), product layer (Doubao Work/Work Companion/phone assistant) to C-end form, already possessing an independent client, phone remote control, OS-level event injection, Feishu permission inheritance and subscription billing. UDOS represents the "research falsifiability" route: with a BFT-lite committee ($n \geq 3f+1$), TransferBundle six-field handoff contract, quality saturation law $mse=a+b/k$, closed-market conservation and three-level evidence system, making multi-Agent organization governance into testable, falsifiable formalized propositions, but currently stopping at CPU prototype, single seed, no real LLM long-chain stage.

This paper's core contribution is a **16-dimension item-by-item comparison master table** (each dimension giving four columns "Doubao status • three states / UDOS status • three evidence levels / who is superior / basis"), plus a bidirectional mutual-learning list and 5 falsifiable predictions. The conclusion: the two should learn from each other rather than replace each other—Doubao's red-team and permission-test engineering can precisely absorb UDOS's BFT-lite, scope least privilege and evidence-grading gates; UDOS's theoretical framework can precisely absorb Doubao's SAEP declarative boundaries, Feishu permission inheritance model and productization launch SOP.

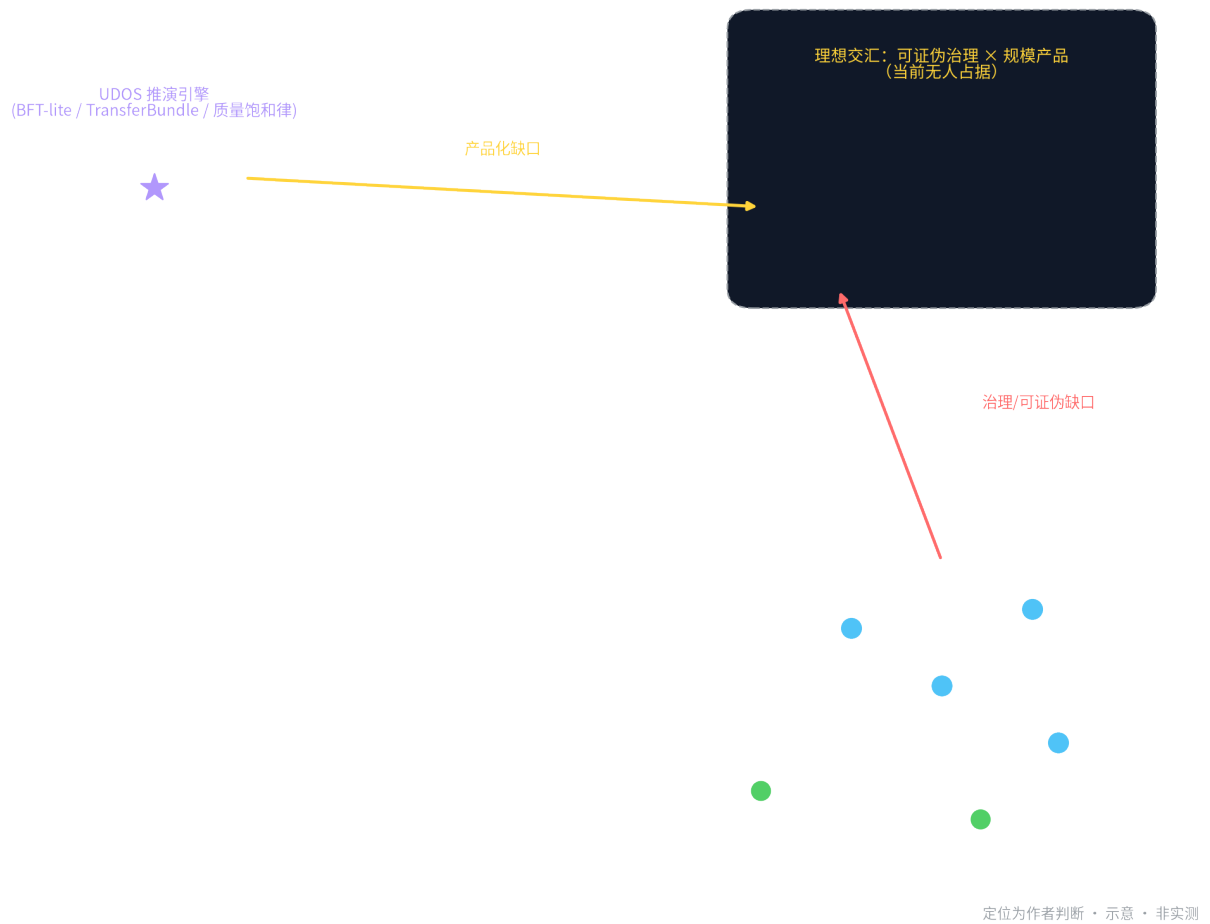
Structured Abstract

- **Background:** in 2026 February Doubao entered the 2.0 stage, June released 2.1 Pro/Turbo, August 25 released the independent "Doubao Work" client, September 15 released Work Companion and Feishu 8.0, a five-layer system covering models to OS Agents has formed. Meanwhile, the UDOS Reasoning Engine on CPU prototypes has sedimented a group of formalized propositions about "how multi-Agents are governed." The two have never been placed together for comparison, this paper first does this.
- **Argument:** industrial productization and falsifiable governance are two **complementary rather than competing** technical routes; maturity not comparable, but in governance rigor UDOS harder, in product thickness Doubao more solid.
- **Evidence:** Doubao side from two official/media specification sheets + this independent online spot check of 14 items (fully confirmed 11, partially confirmed 2, caliber corrected 2, not one falsified); UDOS side from master papers P16–P28 and `reports7/*.json`, all numbers carrying three evidence levels, single seed marked "pilot point estimate."

- **Contribution:** ① 16-dimension comparison master table; ② bidirectional mutual-learning list; ③ 5 falsifiable predictions and scoring time points; ④ one radar chart, one positioning matrix, one mutual-learning roadmap (all qualitative schematic • not measured).

I. Comparison Framework and One-Sentence Conclusion

Placing the two systems on the same chart, first must see clearly they are **fundamentally not in the same maturity quadrant**. Horizontal axis "maturity/real workload and productization," vertical axis "abstraction/formalization and falsifiability." Doubao's all product landings in the lower right—commercial, engineering-leaning; UDOS in the upper left—research prototype, high abstraction. The upper-right ideal intersection quadrant "falsifiable governance × scaled product," **currently occupied by no one**.



This positioning determines the comparison's nature:

- **Not a win-loss question.** Doubao has no obligation to do Byzantine fault tolerance on CPU prototypes, UDOS no obligation to do phone remote control.
- **It is a gap question.** Doubao going from lower right to upper right, lacking "governance/falsifiability"; UDOS going from upper left to upper right, lacking "productization." Two arrows point to the same blank quadrant.
- **It is a mutual-learning question.** Whoever first thickens their own side's engineering/theory, then fills the other side's gap, first occupies the ideal intersection.

Chapter summary: Doubao = industrial productization mature commercial Agent system; UDOS = research falsifiable governance substrate (CPU prototype stage). The comparison's ruler is "governance and falsifiability," not market share, not benchmarks.

II. Sixteen-Dimension Item-by-Item Comparison (Core Master Table)

The table below is the full text's backbone. Each dimension gives four columns: **Doubao status (three-state marked)**, **UDOS status (three evidence levels)**, **who superior**, **basis**. "Who superior" only judged on the same comparable dimension; dimensions with different goals marked "not directly comparable."

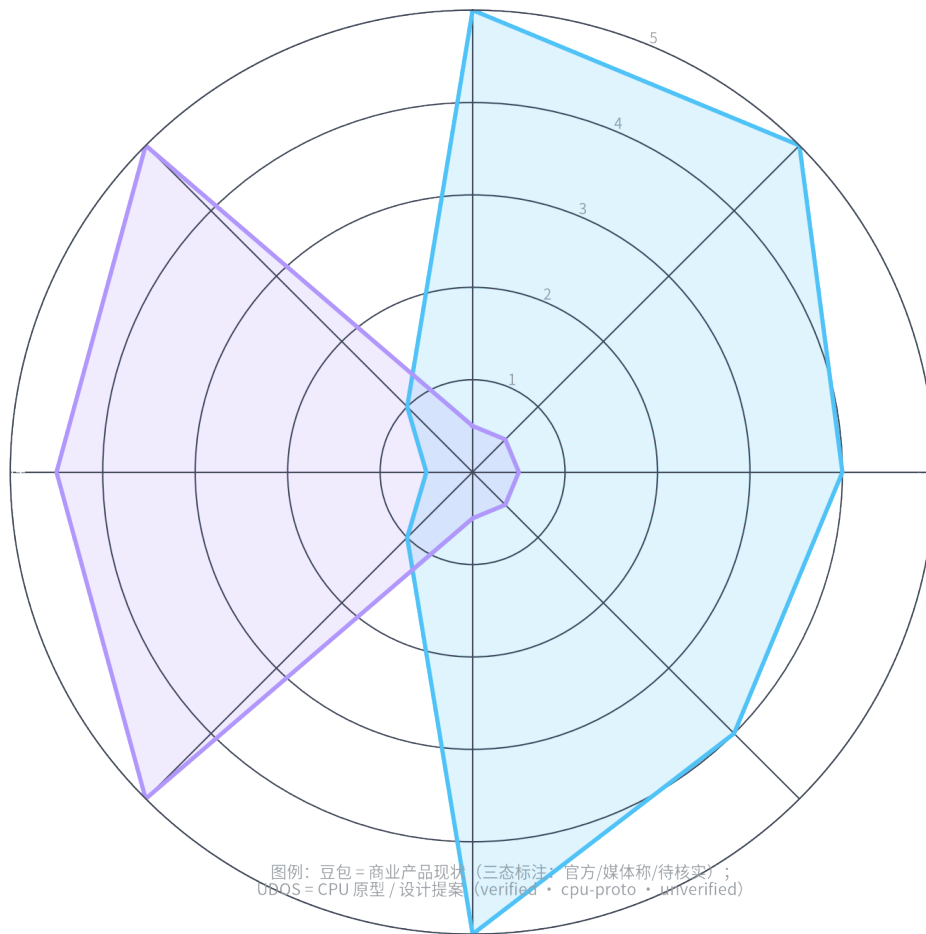
#	Dimension	Doubao status (three states)	UDOS status (three evidence levels)	Who superior	Basis
1	Positioning and goals	Commercial Agent product family facing C-end and enterprise (personal productivity/team/OS-level/enterprise platform), goal real productivity delivery and commercial monetization [official statement]	Research "falsifiable governance substrate," goal making multi-Agent organization governance into testable, falsifiable formalized propositions [unverified • design proposal mainly]	Not directly comparable: commercial delivery Doubao superior, theoretical falsifiability UDOS superior	Doubao five-layer product form; UDOS P16–P28
2	Maturity stage	Scaled commercial/productization: independent client released, phone assistant tech preview, subscription billing 68/200/500 yuan/month [official statement]	CPU prototype/single-seed pilot; most v7.6.0 architecture design proposal • pending implementation [cpu-proto + unverified]	Doubao	2026-08-25 Doubao Work release; reports7
3	Scale and real workload	Real users and real workloads: Doubao 1.8 daily tokens broke 5 trillion [official]; nubia M153 stocked 30,000 units, first batch sold out [media says • caliber corrected to "stocked" not "sold"]; Feishu CLI 767 function points [media says]	No product users; CPU prototype under 24 work orders, 7-person QA committee (2 Byzantine counter-voters) simulation accepted correctly [cpu-proto • single seed]	Doubao	Official release; P26; reports7
4	Model and GUI capability	Self-developed model family: Doubao 2.1 Pro/Turbo (2026-06-24), 2.0 Pro/Lite/Mini/Code (2026-02-14); UI-TARS closed-source version (Mobile Use optimized) GUI substrate [official/media says]; open-source version arXiv:2501.12326 [paper • verified]	No real LLM long chain; CPU prototype small synthetic model (params 967796, torch 2.14.0+cpu) [cpu-proto]	Doubao (model and GUI combat)	UI-TARS paper; reports7/v7_verification.json
5	Protocols and interoperability	MCP/A2A follow-up integration; SAEP Screen Automation operation declarative protocol is Doubao's first creation , from 2026-09-14 30-day public notice [media says • protocol launched by Doubao]; Feishu CLI open-source MIT, 12	Own TransferBundle handoff contract (six fields+scope), not connected to MCP/A2A etc. industrial protocols [six fields verified, scope design proposal]	Doubao (ecosystem interoperability); UDOS (own contract formalization)	36Kr report; P17

#	Dimension	Doubao status (three states)	UDOS status (three evidence levels)	Who superior	Basis
		business domains [media says/press conference retelling]			
6	Multi-Agent orchestration	Work "work squad" multi-specialty Agents, single task split into multi-route sub-Agents in parallel [official statement]; but the division mechanism (research-output-check, main Agent aggregation) official explicit text to verify [to verify]	BFT-lite committee $n \geq 3f+1$, quorum $q=2f+1$; under $n=7, f=2$ ($q=5$) 10 deterministic property tests held [10 tests verified, 24 work orders end-to-end cpu-protocol • single seed]	Doubao (product landing); UDOS (division correctness more formalized)	Ledger component eight; P16/P26
7	Permission and security model	Strictly inherits Feishu permission system "what employees cannot see, Agents cannot obtain"; personal/enterprise data isolation; AI Trust enterprise security system [official caliber]; sensitive operations human review/secondary confirmation [official statement]	scope least privilege "what caller cannot do" by mechanical structure ($s \in \text{scope} \rightarrow \text{scope_violation}$), three invariants (monotonic shrink/over-boundary reject/re-authorization must pass validate) [target verified, current design proposal]	Doubao (landed enterprise permission inheritance); UDOS (least privilege formalization harder)	Ledger component nine; P17
8	Fault tolerance and Byzantine governance	No BFT/Byzantine fault tolerance public design; red-team adversarial testing (privilege escalation/prompt injection/inducement) pre-launch engineering gate [official/spec sheet]; no common-cause fault cutoff	BFT-lite: $n \geq 3f+1$, same-round equivocation whole round discarded, over-f silence returns NO_QUORUM triggering view change [10 deterministic property tests verified]	UDOS	Ledger component ten; P16/P26
9	Handoff contract	Agent-as-Tool (sub-Agents parallel then main Agent reclaims control) present in product; Handoff responsibility transfer no formal contract field specification [to verify]	TransferBundle six fields Goal/Context/Done/Trace/Owner + scope, 26 tests verified [six fields verified, scope design proposal]	UDOS (formalized handoff contract)	P17
10	Quality saturation and diversity handling	No quality saturation governance public design; success rate/function points official or media caliber, no	Ensemble saturation law $mse(k)=a+b/k$: measured $a=0.009252$, $b=-4.23 \times 10^{-5}$, $k>2$ flattens; equal-weight including one bad expert worsens MSE from	UDOS	P20/P9

#	Dimension	Doubao status (three states)	UDOS status (three evidence levels)	Who superior	Basis
		"diversity-quality" relationship formalization [media says/to verify]	about 0.0092 to 0.2131 (about $23\times$); core conclusion: quality first variable is independent information source count D, not headcount N [$k\leq 2$ verified, $k>2$ cpu-proto • single seed]		
11	Settlement and market	Real commercial closed loop: pro subscription 68/200/500, API billing (2.1 Pro input 6 yuan/output 30 yuan per million tokens), terminal sales (M153 3499, NaviX Ultra after national subsidy 5499 starting) [official/media says]; no internal Agent market conservation design	Closed internal market conservation balance_sum=total_paid-slashed, total_paid $\leq$ total_budget, 10 unit tests verified; open market VCG-style design proposal [closed verified, open design proposal]	Doubao (real commercial closed loop); UDOS (settlement conservation invariant)	Ledger; P25
12	Evidence and reproducibility	Success rate/function points/skill counts etc. official releases or media retellings, not independently reproducible ; spec sheet itself lists 14 "to verify" items [official/media says/to verify three states]	Conclusions graded by verified/cpu-proto/cpu-proxy/unverified; cpu-proto numbers in reports7/*.json rerunnable; single seed uniformly marked "pilot point estimate," before submission must add ≥ 30 seeds + 95% CI [system itself]	UDOS (evidence discipline)	Ledger part five; v7.5.0 P12
13	Openness	UI-TARS open source (github.com/bytedance/UI-TARS, arXiv: 2501.12326) [official/paper]; Feishu CLI open-source MIT; product itself closed source	CPU prototype code and reports7 internally rerunnable; not externally open sourced [cpu-proto]	Tie: Doubao has real open-source model and CLI; UDOS has rerunnable prototype but not publicly released	arXiv; reports7
14	Data flywheel and memory	Single-role conversation memory + team long-term memory (one person teaches, whole team can use) + behavior personalization memory under user authorization [official statement]; data flywheel relies on real user usage	No real user data flywheel; memory structured Trace, retained along handoff chain [design]	Doubao (real data flywheel)	Ledger component seven; P17
15		Cloud PC/Windows virtual desktop/phone remote control; system-	Pure CPU prototype; Warp-Cortex singleton weight sharing presses memory complexity		Ledger; P28/P22

#	Dimension	Doubao status (three states)	UDOS status (three evidence levels)	Who superior	Basis
	Edge-side and cloud infrastructure	level screenshot+event injection (WindowManagerService/inject_events only vendor preinstalled); AI cloud-native (GPU core) [official/media says]; partner vendor OS-layer preinstallation (nubia/Nubia NaviX Ultra)	$O(N \cdot L) \rightarrow O(1) + O(N \cdot k)$ (paper reports: single 4090 about 100 concurrent, 2.2GB VRAM) [paper reports/cpu-proto]	Doubao (end-to-end edge engineering)	
16	Governance philosophy	Engineering governance: human review gates/red teams/permission tests/canary rollback, relying on process and human checks; fast-iterating commercial system	Formalized governance: structural changes must carry test assertions, missing assertions mechanically rejected, evidence-grading gates, relying on structure and invariants [design proposal • pending implementation]	Different philosophies, each has strengths	Ledger SOP; P19

Compressing these 16 dimensions into one radar chart (8 representative dimensions, qualitative scoring 0–5), the shape immediately clear: Doubao bulges on the first 5 axes (product/engineering/protocol/permission/commerce), UDOS bulges on the last 3 axes (falsifiability/BFT/rerunnable).



作者基于公开资料的定性评估 · 非实测 · 2026-09-21

Chapter summary: among 16 dimensions, Doubao clearly superior on 6 dimensions "scale workload, model GUI, protocol ecosystem, commercial closed loop, data flywheel, edge engineering"; UDOS clearly superior on 4 dimensions "Byzantine governance, handoff contract, quality saturation, evidence reproducibility"; remaining dimensions each have merits or different goals not comparable. The radar chart's two bulge directions precisely what the two routes each do thoroughly.

III. The Doubao System's Advantages: The Real Thickness of Industrial Productization

Doubao's advantage is not some benchmark, but **a whole engineering chain from model to OS already running through on real users.**

1. **Real product and user scale.** Doubao 1.8 official caliber daily tokens broke 5 trillion [official]; Doubao Work 2026-08-25 released independent client, pro 68/200/500 yuan/month three-tier subscription online [official]; phone assistant first-generation nubia M153 stocked 30,000, first batch sold out [media says]. This is what UDOS entirely lacks.

2. **End-to-end engineering.** Local+cloud PC dual mode, Windows virtual desktop, phone remote controlling computer, system-level screenshot+event injection—from "helping you write documents" all the way to "clicking the screen for you." This GUI operation chain (screenshot perception→reasoning decomposition→element localization→event injection→reflective correction) is genuinely landed hard engineering.
3. **Protocol ecosystem.** Besides MCP/A2A follow-up integration, **SAEP (Screen Automation declarative protocol) is Doubao's first creation:** third-party apps autonomously declare allowing/refusing/limiting AI automated operations within this app, from 2026-09-14 30-day public notice, with traceable operation logs [media says]. This is a declarative design about "AI operation boundaries," intent one level higher than "can it click."
4. **Feishu permission inheritance and data isolation.** Doubao Work logs in via Feishu account, inheriting enterprise context within **permission scope** (group chats/docs/meeting minutes/schedules), and strictly guards "what employees cannot see, Agents cannot obtain"; personal/enterprise data isolation [official caliber]. This is enterprise Agent's hardest foundation.
5. **Skill connector ecosystem and model family iteration speed.** Skill center over 200 skills/connectors [official/media]; model family from 2024-05 first release to 2026-06's 2.1 Pro/Turbo, iteration rhythm quarterly; Coze/TRAE/AgentKit/HiAgent/ArkClaw/AI Trust fill the whole low-code to high-code development chain.
6. **Commercial closed loop.** Subscription, API billing, terminal sales three lines parallel, already self-sustaining.

Chapter summary: Doubao's moat is "real"—real users, real workloads, real payment, real edge. It proves how far a commercial Agent system can go.

IV. The Doubao System's Shortcomings and Black Box: Falsifiability Absent

Shortcomings must be written evidence-grounded, not blackening. Problems concentrated in **"numbers spoken out, but others cannot reproduce; division spoken about, but not written in black and white; governance relies on process, but no formalized fallback."**

1. **Key numbers official or media caliber, not independently reproducible.** Feishu CLI success rate 78%→95%, function points 247→767, skills over 200, daily tokens 5 trillion—these are releases/press conferences/media retellings [media says], outsiders cannot obtain test sets, cannot reproduce. The spec sheet itself also lists **14 "to verify" items** (e.g. Turbo/Pro concurrency and context length differences not public, skill review mechanism not public, etc.).
2. **Multi-Agent division mechanism not officially documented.** "Work squad," "research-output-check, main Agent aggregation" currently product narrative [to verify], no public white paper explaining clearly how sub-Agents divide labor, avoid duplicate work, converge conflicts.
3. **No BFT/common-cause fault cutoff.** When multiple Agents work simultaneously, if all share the same upstream error (common-cause fault), Doubao system's public materials have no "whole round discard/change view" type fault-tolerant design; its security relies on **red-team adversarial testing + human review gates + permission tests** engineering gates, "pre-launch checks," not "runtime fault tolerance."
4. **No quality saturation governance, no settlement conservation.** No formalized conclusions like "after adding the kth expert marginal returns decline, even equal-weight including a bad expert

worsens quality $23\times$ "; how many people multi-Agent invests, whether over-saturated, relies on experience not laws.

5. **No evidence-grading gates.** Conclusions not divided verified/cpu-proto/unverified, single-seed results not automatically marked "pilot point estimate," readers cannot see at a glance which number stands.

Chapter summary: Doubao's black box is not "hiding bad things," but "**good numbers lack a reproducible evidence skeleton.**" Its red-team and permission-test engineering done solidly, but lacks the layer sedimenting these engineering into formalized propositions.

V. The UDOS Reasoning Engine's Advantages: A Falsifiable Governance Substrate

UDOS's value precisely in the layer Doubao lacks—making "how multi-Agents governed" into **testable, falsifiable** propositions. The following numbers all carry three evidence levels, not exaggerated.

1. **Falsifiable propositions and three evidence levels.** All conclusions speak graded by verified/cpu-proto/cpu-proxy/unverified; single-seed results uniformly marked "pilot point estimate," before submission must add ≥ 30 seeds + 95% CI. This is a discipline letting readers judge "how hard this conclusion is."
2. **BFT-lite Byzantine fault tolerance.** Committee $n \geq 3f+1$, quorum $q = 2f+1$; same-round equivocation whole round discarded; over- f silence returns NO_QUORUM triggering view change. Under $n=7, f=2$ ($q=5$), **10 deterministic property tests** hold "honest supermajority stops/Byzantine cannot force-stop/silence transitions view"; 7-person QA committee under 2 Byzantine counter-voter members accepted all 24 work orders correctly. **10 tests verified; 24 work orders end-to-end cpu-proto (single seed).**
3. **TransferBundle handoff contract.** Handoff package six fields Goal/Context/Done/Trace/Owner, 26 tests verified; v7.6.0 adds seventh field scope (stages current owner allowed to operate), advance automatically shrinks, handoff re-authorizes, over-boundary rejects (scope_violation). **Six fields verified; scope design proposal • pending implementation.**
4. **Quality saturation law.** Ensemble saturation law $mse(k) = a + b/k$; CPU prototype measured $a=0.009252, b=-4.23 \times 10^{-5}$, after $k>2$ flattens; equal-weight including one bad expert worsens MSE from about 0.0092 to **0.2131 (about $23\times$)**. Core conclusion: quality first variable is independent information source count D , not headcount N . **$k \leq 2$ verified, $k>2$ cpu-proto (single seed).**
5. **Topology and memory complexity.** Layered trees press each node fan-in to constant 9, communication rounds to 14 (cpu-proto); Warp-Cortex singleton weight sharing presses memory complexity from $O(N \cdot L)$ to $O(1)+O(N \cdot k)$.
6. **Market conservation and review gates.** Closed internal market conservation $balance_sum = total_paid - slashed, total_paid \leq total_budget$, 10 unit tests verified; before structural change merge mandatory "independent contract test assertions + key regression all green + CHANGELOG marking" three-piece set, missing assertions mechanically rejected (design proposal).

Chapter summary: UDOS splits "how multi-Agent organizations governed" into a group of test-runnable, falsifiable propositions. Its numbers not pretty but hard—because each number followed by evidence grade and rerun path.

VI. The UDOS Reasoning Engine's Shortcomings: The Research Prototype's Ceiling

Honestly, UDOS currently is **not a usable product**. Shortcomings likewise must be written clearly, not deified.

1. **CPU prototype / single seed / no real LLM long chain.** All cpu-proto numbers run on CPU, single seed, using 96,000-parameter-class synthetic models, not connected to real large models' long-chain reasoning. The $23\times$ worsening, $k>2$ flattening conclusions currently all "pilot point estimates."
2. **No product users, no real workloads.** Not one real user using, much less real traffic, real enterprise permission scenarios; 24 work orders simulated work orders, not online incidents.
3. **No GUI/edge-side/permission system.** No cloud PC, no phone remote control, no OS-level event injection, nor a ready enterprise identity permission system like Feishu to inherit.
4. **No protocol ecosystem and commercial landing.** Not connected to MCP/A2A/SAEP, no subscription billing, no customers.
5. **Most new architecture still design proposals.** Three-level typed finality, scope field, review gates, open market VCG—these v7.6.0 new things overwhelmingly marked [RESULT NEEDED], no measured gains yet.

Chapter summary: UDOS's ceiling is "not yet entered the real world." Its proposed propositions good, but to reproduce on real LLMs, real users, real workloads, only then counts.

VII. Bidirectional Mutual-Learning List

This is the full text's most practical table: what Doubao can borrow from UDOS, what UDOS can borrow from Doubao. Both directions not "copying," but "connecting the piece the other side already thought clearly to their own engineering/theory."

Direction	Learn what	Land on own existing capability	Expected gain
Doubao ← UDOS	BFT-lite committee stop	Upgrading "work squad" multi-sub-Agent stop/convergence from main Agent self-awareness to $n \geq 3f+1$ majority decision, same-round errors whole round discarded	Common-cause faults no longer wrong all the way
Doubao ← UDOS	scope least privilege	Upgrading "if employees cannot see, Agents cannot obtain" from permission test cases to mechanical invariants (over-boundary reject, not silently pass)	Least privilege relies on structure not human review
Doubao ← UDOS	Evidence-grading gates	Marking release-style success/function points per verified/cpu-proto/unverified, single seed automatically marked "pilot point estimate"	Key numbers reproducible, traceable
Doubao ← UDOS	Quality saturation law	Using $mse=a+b/k$ to guide "how many sub-Agents a task should dispatch, when to stop adding people"	Avoid multi-Agent over-saturation instead degrading quality
Doubao ← UDOS	Review gates	Engineering red-team/permission tests: before structural changes (schema/contract field changes) merge mandatory test assertions, missing assertions mechanically rejected	Launch gate from "human feels okay" to "tests green only"
UDOS ← Doubao	SAEP declarative operation boundaries	Making "whether AI can operate some external app" into a third-party declarable, traceable protocol layer, not hardcoded	From simulation toward real application ecosystem
UDOS ← Doubao	Feishu permission inheritance model	Enterprise Agent not starting another permission system, directly inheriting existing identity system and data isolation boundaries	Enterprise landing permission foundation
UDOS ← Doubao	Cloud PC/edge engineering	Filling GUI operations (screenshot→localization→injection) and cloud phone/virtual desktop execution environment	From text Agent toward able to operate systems
UDOS ← Doubao	Skill review and canary rollback	Pre-launch red-team adversarial + directed co-creation canary + one-click rollback (close Agent/remove connector/retreat version)	Research systems also safely iterate
UDOS ← Doubao	Productization launch SOP	Borrowing eight-stage×four-role SOP to release, canary, operate research prototypes at engineering rhythm	From prototype toward usable product



互学清单为作者建议 · 非厂商路线承诺 · 示意

Chapter summary: the "governance skeleton" Doubao lacks is precisely UDOS's ready one; the "product flesh" UDOS lacks is precisely Doubao's ready one. The ideal intersection quadrant not imagined—it is what this mutual-learning table looks like after landing.

VIII. Five Falsifiable Predictions

The following 5 all written in "overturnable by future facts" form, scoring time uniformly **2027-09-21** (one year after this paper's publication).

#	Prediction	If holds (which of this paper's judgments falsified)	If not holds (maintain this paper's judgment)	Scoring time
1	Doubao will publicly release multi-Agent division (research-output-check/main Agent aggregation) mechanism white paper and reproducible success rate	Then the "multi-Agent division mechanism not officially documented" shortcoming falsified, Doubao catches up to UDOS's formalized clarity in this dimension	Still no white paper—maintain "division not documented" judgment	2027-09-21
2	Doubao will introduce BFT-lite-style committee stop/common-cause fault cutoff and publicly test	Then its fault tolerance upgrades from "red-team engineering" to "formalized fault tolerance," dimension 8 advantage changes hands	Still relies on human review gates, no formalized fault tolerance—maintain	2027-09-21
3	UDOS will on real LLM long chains (≥ 3 real models, ≥ 30 seeds, 95% CI) reproduce BFT-lite stop correctness and quality saturation law $mse=a+b/k$	Then cpu-proto single-point estimate upgrades to verified, research value genuinely lands, "prototype limitation" shortcoming partially falsified	Still stops at CPU synthetic models—maintain "CPU prototype/single seed" judgment	2027-09-21
4	Doubao will make "if employees cannot see, Agents cannot obtain" permission inheritance public as reproducible formalized assertions/test suites (rather than official caliber)	Then its reproducibility shortcoming improves, evidence discipline catches up to UDOS	Still only official caliber, no test suites—maintain	2027-09-21
5	By scoring time, an ideal intersection system "falsifiable governance $\times$ scaled product" will appear (both reaching Doubao-level real workload and carrying UDOS-level evidence-grading gates)	Then this comparison framework positioning matrix's "upper right occupied by no one" judgment overturned	Still occupied by no one—maintain two systems in two quadrants judgment	2027-09-21

Chapter summary: a comparison paper's value is not in saying who is strong now, but in **writing propositions testable by the future**. Looking back at these 5 one year later, whichever overturned, one more point of understanding than now.

IX. Failure Boundaries

- Material cutoff:** this comparison based on public materials and two spec sheets as of **2026-09-21**; after this any Doubao or UDOS updates may make conclusions stale.
- Doubao iterates extremely fast:** Doubao a fast-iterating commercial system, versions, pricing, function points change weekly/monthly, this paper's numbers (68/200/500, 767, 78%→95%, 200+, 2026-08-25 etc.) only represent that time's caliber, not current or future.
- UDOS mostly single-seed pilots:** UDOS side numbers mostly cpu-proto single-seed pilot point estimates, no multi-seed and 95% CI, extrapolation cautious; most v7.6.0 new architecture design proposal • pending implementation.

4. **Qualitative scoring not measured:** radar chart and positioning matrix landings author's **qualitative assessment** based on public materials, not either side's measured benchmarks or official scores.
 5. **Not constituting procurement advice:** this paper technical contrast and research notes, not constituting procurement, investment or selection advice for any product; Doubao product names only describe public positioning, not endorsing vendors.
-

References

1. Doubao Agent system "Architecture and Design Specification Sheet" (light-rendered version, 14 tables, user-uploaded attachment, 2026-09).
 2. "Doubao Agent / Intelligent Agent / Super Intelligent Agent—Architecture and System • Design Specification Sheet • SOP" (user-uploaded attachment, containing three-state source marking, 2026-09).
 3. ByteDance Seed & Tsinghua. UI-TARS: Pioneering Automated GUI Interaction with Native Agents. arXiv:2501.12326, 2025-01-22. <https://arxiv.org/abs/2501.12326>
 4. Economic Information Daily: Doubao large model (Skylark) first release, 2024-05-15. jjckb.cn/2024-05/15/c_1310774888.htm
 5. CNR: Doubao large model enters 2.0 stage (Pro/Lite/Mini+Code), 2026-02-14. tech.cnr.cn
 6. China Science and Technology Network/Jingbao News/The Paper: Doubao pro version online, connected to 2.1 Pro/Turbo, 2026-06-24.
 7. Guangming Web: Doubao Work independent client released, 2026-08-25; skills and connectors over 200, 2026-08-21. [economy.gmw.cn / it.gmw.cn](http://economy.gmw.cn/it.gmw.cn)
 8. 36Kr: SAEP Screen Automation declarative protocol 30-day public notice, 2026-09-14; Xie Xin on Feishu CLI 247→767 / 78%→95% / speed +39%, 2026-09-15.
 9. China Daily/aily official help center: Feishu aily renamed "Doubao Work Companion" (renamed 2026-08-14, 2026-09-15 conference release), "domestic first team Agent."
 10. Jiemian/ZOL: first-generation nubia M153 3499 yuan, stocked 30,000, first batch sold out, 2025-12-01.
 11. 36Kr/Sina: Nubia NaviX Ultra official price 5999 yuan, after national subsidy 5499 starting, 2026-09.
 12. UDOS Reasoning Engine master papers P16–P28 and `reports7/*.json` (BFT-lite stop decision, TransferBundle, quality saturation law, topology complexity, market conservation, three evidence levels), v7.5.0/v7.6.0/v7.7.x read-only retelling.
-

UDOS v7.7.3 • Doubao two papers • lower paper (comparison paper) • 2026-09-21